

OpenAI Function Calling

Notes:

- LLM's don't always produce the same results. The results you see in this notebook may differ from the results you see in the video.
- Notebooks results are temporary. Download the notebooks to your local machine if you wish to save your results.

```
In [1]: import os
import openai

from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file
openai.api_key = os.environ['OPENAI_API_KEY']
```

```
In [2]: import json

# Example dummy function hard coded to return the same weather
# In production, this could be your backend API or an external API
def get_current_weather(location, unit="fahrenheit"):
    """Get the current weather in a given location"""
    weather_info = {
        "location": location,
        "temperature": "72",
        "unit": unit,
        "forecast": ["sunny", "windy"],
    }
    return json.dumps(weather_info)
```

```
In [3]: # define a function
functions = [
    {
        "name": "get_current_weather",
        "description": "Get the current weather in a given location",
        "parameters": {
            "type": "object",
            "properties": {
                "location": {
                    "type": "string",
                    "description": "The city and state, e.g. San Francisco,"
                },
                "unit": {"type": "string", "enum": ["celsius", "fahrenheit"]}
            },
            "required": ["location"],
        },
    },
]
```

```
In [4]: messages = [
        {
            "role": "user",
            "content": "What's the weather like in Boston?"
        }
    ]
```

```
In [5]: import openai
```

```
In [6]: response = openai.ChatCompletion.create(
        model="gpt-3.5-turbo-0613",
        messages=messages,
        functions=functions
    )
```

```
In [7]: print(response)

{
  "id": "chatcmpl-8xGUYjPUJUKuqcvKtIKCCP9sgA3iy",
  "object": "chat.completion",
  "created": 1709135438,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": null,
        "function_call": {
          "name": "get_current_weather",
          "arguments": "{\n  \"location\": \"Boston, MA\"\n}"
        }
      },
      "logprobs": null,
      "finish_reason": "function_call"
    }
  ],
  "usage": {
    "prompt_tokens": 82,
    "completion_tokens": 18,
    "total_tokens": 100
  },
  "system_fingerprint": null
}
```

```
In [8]: response_message = response["choices"][0]["message"]
```

```
In [9]: response_message
```

```
Out[9]: <OpenAIObject at 0x7f062c0f6270> JSON: {
  "role": "assistant",
  "content": null,
  "function_call": {
    "name": "get_current_weather",
    "arguments": "{\n  \"location\": \"Boston, MA\"\n}"
  }
}
```

```
In [10]: response_message["content"]
```

```
In [12]: response_message["function_call"]
```

```
Out[12]: <OpenAIObject at 0x7f062c0fe270> JSON: {
  "name": "get_current_weather",
  "arguments": "{\n  \"location\": \"Boston, MA\"\n}"
}
```

```
In [13]: json.loads(response_message["function_call"]["arguments"])
```

```
Out[13]: {'location': 'Boston, MA'}
```

```
In [14]: args = json.loads(response_message["function_call"]["arguments"])
```

```
In [15]: get_current_weather(args)
```

```
Out[15]: '{"location": {"location": "Boston, MA"}, "temperature": "72", "unit": "fahrenheit", "forecast": ["sunny", "windy"]}'
```

```
In [16]: messages = [
  {
    "role": "user",
    "content": "hi!",
  }
]
```

```
In [17]: response = openai.ChatCompletion.create(
  model="gpt-3.5-turbo-0613",
  messages=messages,
  functions=functions,
)
```

```
In [18]: print(response)
```

```

{
  "id": "chatcmpl-8xGW9oLklc9pCiiAMsggu72I2lfXQ",
  "object": "chat.completion",
  "created": 1709135537,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Hello! How can I assist you today?"
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 76,
    "completion_tokens": 10,
    "total_tokens": 86
  },
  "system_fingerprint": null
}

```

```

In [19]: messages = [
          {
            "role": "user",
            "content": "hi!",
          }
        ]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    functions=functions,
    function_call="auto",
)
print(response)

```

```

{
  "id": "chatcmpl-8xGXYeAK0MX8ICHBnFtA9SvmRSKiK",
  "object": "chat.completion",
  "created": 1709135624,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Hello! How can I assist you today?"
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 76,
    "completion_tokens": 10,
    "total_tokens": 86
  },
  "system_fingerprint": null
}

```

```

In [20]: messages = [
    {
      "role": "user",
      "content": "hi!",
    }
  ]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    functions=functions,
    function_call="none",
)
print(response)

```

```

{
  "id": "chatcmpl-8xGYDTaSK3vQMaxnY5SjalDww5c5c",
  "object": "chat.completion",
  "created": 1709135665,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Hello! How can I assist you today?"
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 77,
    "completion_tokens": 9,
    "total_tokens": 86
  },
  "system_fingerprint": null
}

```

```

In [22]: messages = [
    {
        "role": "user",
        "content": "What's the weather in Boston?",
    }
]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    functions=functions,
    function_call="none",
)
print(response)

```

```

{
  "id": "chatcmpl-8xGYvmZJ67sU7KAAX3Y7bLsCLCpka",
  "object": "chat.completion",
  "created": 1709135709,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Please give me a moment. I will check the weather in Bos
ton for you."
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 82,
    "completion_tokens": 16,
    "total_tokens": 98
  },
  "system_fingerprint": null
}

```

```

In [23]: messages = [
    {
        "role": "user",
        "content": "hi!",
    }
]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    functions=functions,
    function_call={"name": "get_current_weather"},
)
print(response)

```

```

{
  "id": "chatcmpl-8xGZ1XUwW9bZM60aL23JWYX5s41Xo",
  "object": "chat.completion",
  "created": 1709135715,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": null,
        "function_call": {
          "name": "get_current_weather",
          "arguments": "{\n  \"location\": \"San Francisco, CA\"\n}"
        }
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 83,
    "completion_tokens": 12,
    "total_tokens": 95
  },
  "system_fingerprint": null
}

```

```

In [29]: # increased prompt_tokens due to function calling
messages = [
    {
        "role": "user",
        "content": "What's the weather like in Boston!",
    }
]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    functions=functions,
    function_call={"name": "get_current_weather"},
)
print(response)

```



```

{
  "id": "chatcmpl-8xGcVzxGuhHemtVBj8CekYwfKKsok",
  "object": "chat.completion",
  "created": 1709135931,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": null,
        "function_call": {
          "name": "get_current_weather",
          "arguments": "{\n  \"location\": \"Boston, MA\"\n}"
        }
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 89,
    "completion_tokens": 11,
    "total_tokens": 100
  },
  "system_fingerprint": null
}

```

```

In [26]: # Decreased prompt_tokens due to function calling
messages = [
    {
        "role": "user",
        "content": "What's the weather like in Boston!",
    }
]
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo-0613",
    messages=messages,
    # functions=functions,
    # function_call={"name": "get_current_weather"},
)
print(response)

```

```
{
  "id": "chatcmpl-8xGbBNCx9ZwUVFJvzrE1WA2mD5fjK",
  "object": "chat.completion",
  "created": 1709135849,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "I am sorry, but as an AI, I do not have real-time data. However, you can check the weather in Boston by referring to local weather websites or using weather forecasting applications on your smartphone or computer."
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 15,
    "completion_tokens": 43,
    "total_tokens": 58
  },
  "system_fingerprint": null
}
```

```
In [30]: messages.append(response["choices"][0]["message"])
```

```
In [31]: args = json.loads(response["choices"][0]["message"]['function_call']['arguments'])
         observation = get_current_weather(args)
```

```
In [32]: observation
```

```
Out[32]: '{"location": {"location": "Boston, MA"}, "temperature": "72", "unit": "fahrenheit", "forecast": ["sunny", "windy"]}'
```

```
In [33]: messages.append(
         {
             "role": "function",
             "name": "get_current_weather",
             "content": observation,
         }
     )
```

```
In [34]: response = openai.ChatCompletion.create(
         model="gpt-3.5-turbo-0613",
         messages=messages,
     )
     print(response)
```

```
{
  "id": "chatcmpl-8xGd0ogVPJ0mefXWqpVjpdINdigSj",
  "object": "chat.completion",
  "created": 1709135986,
  "model": "gpt-3.5-turbo-0613",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "The weather in Boston is currently 72\u00b0F (22\u00b0C)
with sunny skies and windy conditions."
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 77,
    "completion_tokens": 20,
    "total_tokens": 97
  },
  "system_fingerprint": null
}
```

In []:

In []: